

# Beyond Pass@1: Reliability and Token-Cost Effects of Quantized Reasoning Models under a Pinned Serving Stack

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## Abstract

Post-training quantization is a common serving default for reasoning language models, but pass@1 on a single seed can hide changes in trace length, termination behavior, and sample-level reliability. We pin one serving stack—vLLM 0.7.0 [10] eager execution on an NVIDIA A100-80GB, with FP8 checkpoints executed as Marlin W8A16 [5] rather than native W8A8—and vary only the weight checkpoint among BF16, FP8, AWQ-4, and GPTQ-4 for DeepSeek-R1-Distill-Qwen-7B and DeepSeek-R1-Distill-Llama-8B. The grid has 88 cells and 56,408 completions on MATH-500 [8, 13] (5 seeds), GSM8K [2] (3 seeds), and GPQA-Diamond [16] (3 seeds).

On MATH-500, FP8-BF16 pass@1 differences are +0.40 and +0.28 percentage points (pp). Problem-clustered bootstrap 95% intervals include zero; a  $\pm 1$  pp TOST equivalence test is *not* passed. Four-bit effects are architecture- and task-dependent: Llama AWQ-4 loses 2.76 pp on MATH-500 (95% CI  $[-4.16, -1.44]$ ,  $p < 0.001$ ) and 1.57 pp on GSM8K; Qwen AWQ-4 loses 5.56 pp on GPQA-Diamond (95% CI  $[-9.60, -1.52]$ ,  $p = 0.007$ ). Majority-vote McNemar tests on maj@5 remain non-significant and do not answer the pass@1 question. Mean MATH-500 length rises 6.3–6.8% for Qwen 4-bit (paired token CIs exclude zero); the increase is concentrated on items where BF16 is correct and the quantized run is not. Identical-word loops are rare but nonzero (25/56,408); none of the rows hit the 32,768 token cap after re-encoding, yet 209 completions sit at  $\geq 32,500$  tokens, and Qwen 4-bit near-cap counts on MATH-500 are about  $1.7\times$  BF16. Using recovered answer strings, gold-free modal agreement is an observable abstention signal: 5/5 consensus has very low selective error on MATH-500 but costs five generations per query, and Llama AWQ-4 reduces 5/5 coverage by 6.0 pp versus BF16 (95% paired CI  $[-9.4, -2.6]$ ). On a frozen 100-problem MATH-500 serving subset, batched Qwen-7B FP8 decode is +18.7% faster than matched BF16 (824 vs. 694 tok/s) and –19.8% lower in measured Cost-of-Pass at a \$1.50/A100-hour scenario. Four-bit throughput is concurrency-dependent: every 4-bit cell is slower than BF16 in single-stream decode, while batched Llama GPTQ-4 reaches 786 tok/s. Peak allocated VRAM stays  $\sim 55$ –57 GB across formats under `gpu_memory_utilization=0.75` and does not measure weight-footprint savings.

**Keywords:** Reasoning language models, post-training quantization, pinned serving stack, token inflation, selective prediction, Cost-of-Pass.

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# 1 Introduction

Reasoning models such as DeepSeek-R1 [7] emit long traces before a final answer. Weight-only post-training quantization (FP8, AWQ-4 [14], GPTQ-4 [4]) is therefore a common serving choice. Isolated pass@1 numbers hide whether compression also changes trace length [12, 15], termination, or the cost of a correct answer [3]. Serving-stack details—engine version, decoding flags, CUDA graphs—can move reasoning scores as much as model changes [9]. This paper *pins* one stack rather than shifting it: we do not run a factorial vLLM 0.7.0 vs. 0.8.5 experiment.

We ask four questions under that pin:

- **RQ1.** Does quantization change *pass@1* relative to matched BF16, with problem-clustered uncertainty?
- **RQ2.** Do identical-word loops and near-budget terminations occur, and do they differ by format?
- **RQ3.** What can be said about multi-sample reliability without gold labels at serve time?
- **RQ4.** How do measured decode throughput and GPU-seconds per query change Cost-of-Pass relative to a shared-throughput token proxy?

**Contributions.** (1) A released, format-matched, stack-pinned 88-cell grid (56,408 completions) with per-cell JSON. (2) Problem-clustered pass@1 inference showing FP8–BF16 intervals that include zero, a statistically detectable Llama AWQ-4 drop on MATH-500/GSM8K, and a Qwen AWQ-4 drop on GPQA-Diamond—effects that maj@5 McNemar tests miss. (3) Evidence that 4-bit token inflation is real for Qwen on the full MATH-500 grid, is *not* an artifact of a 200-item even-index subset, and is concentrated on format-induced failures rather than uniformly longer correct traces. (4) Gold-free modal-answer agreement on MATH-500: higher observable agreement is associated with lower selective error, while 4-bit AWQ reduces strict-consensus coverage in some configurations even when selective error among retained predictions stays low. (5) Measured serving on the same pinned stack (48 task-realistic runs + 8 fixed-token microbenchmarks): Cost-of-Pass uses campaign MATH-500 pass@1 and measured GPU-seconds rather than a shared 65 tok/s assumption.

## 2 Related Work

GPTQ [4] and AWQ [14] were originally validated mainly on short-form benchmarks; SmoothQuant [18] addresses activation quantization, which we do not run. Liu et al. [15] evaluate DeepSeek-R1 distillations across weight, activation, and KV quantization and report near-lossless W8/W4A16 in many settings, with *no* systematic output lengthening. Lian et al. [12] report the opposite: low-bit PTQ can inflate chain-of-thought length. Li et al. [11] localize 4-bit math failures to early process errors. Alimaskina et al. [1] study loops and delayed commitment under extreme low-bit inference. Our delta is a single pinned vLLM 0.7.0 eager path [10] with five MATH-500 seeds and paired item-level tests, not a new compressor.

Hochlehnert et al. [9] show that seeds, prompts, decoding, and software move reasoning scores; we freeze those factors. Zhong et al. [19] study uncertainty calibration of quantized LLMs using model probabilities—a quantity we do not log. Self-consistency [17] and later unsupervised estimators [20] treat sample agreement as confidence; selective classification [6] supplies the risk-coverage language. Erol et al. [3] define Cost-of-Pass as expected monetary cost per correct solution; we report it twice: a historical shared-65 tok/s token proxy, and a measured GPU-second version on a frozen MATH-500 serving subset.

## 3 Experimental Methodology

### 3.1 Models and precision formats

We evaluate DeepSeek-R1-Distill-Qwen-7B and DeepSeek-R1-Distill-Llama-8B [7]. Formats: uncompressed **BF16**; **FP8** checkpoints served on A100 via vLLM’s Marlin **W8A16** weight-only fallback (not native W8A8 tensor-core FP8); **AWQ-4** (group size 128, float16); **GPTQ-4** (group size 128, Marlin). Hugging Face IDs and revisions are listed in the appendix. AWQ-4 checkpoints are third-party community uploads; “Llama AWQ-4 is weak” is a property of those artifacts, not a proof about AWQ as a method.

### 3.2 Benchmarks and protocol

MATH-500 [8, 13] ( $n = 500$ , seeds 42–46): 20,000 completions. GSM8K [2] ( $n = 1,319$ , seeds 42–44) and GPQA-Diamond [16] ( $n = 198$ , seeds 42–44): breadth. Decoding: zero-shot <think> prefix (appendix),  $T = 0.6$ , top- $p = 0.95$ , repetition penalty 1.0 (deliberately not used to hide loops), max new tokens 32,768. Total: 56,408 completions in 88 cells. MATH and GSM8K are widely used; contamination likely inflates absolute scores but largely cancels in within-model format contrasts.

### 3.3 Hardware and serving stack

PARAM Rudra HPC (IIT Patna / NSM), NVIDIA A100-PCIE-80GB. Publication runs use the official QRM inference.py path [15] with `gpu_memory_utilization=0.75`. Engine pin: `vLLM==0.7.0`, eager execution. Source files under `configs/models/` are *not* the campaign launcher; they differ in `max_model_len` and KV-cache defaults and should not be read as the effective stack.

### 3.4 Measured serving

After the 56k accuracy campaign we timed the same eight checkpoints on A100-80GB under vLLM 0.7.0 eager (protocol: docs/MEASURED\_SERVING\_PROTOCOL.md). Workload: a frozen 100-problem MATH-500 list (20 problems per difficulty label). Prompts are stored as `full_prompt` in `results/measured_serving/input_subset.json`; the sidecar `problem/solution` metadata in that JSON is misaligned with the prompts and is not used. Condition A: sequential single-request decode of the first 20 prompts of that list (level counts 5/7/3/3/2, not a re-stratified 4-per-level draw),  $R = 3$ . Condition B: one `llm.generate` call over all 100 prompts (continuous batching; labeled  $C = 8$  in the protocol, but `max_num_seqs` was not pinned to 8). Secondary microbenchmark: 10 prompts forced to exactly 512 tokens (`ignore_eos`). Three warmup queries are discarded. Sampling matches the campaign ( $T = 0.6$ , top- $p = 0.95$ ) with seed 20260816. Repetitions share that seed, so output token counts are identical across reps; the  $R = 3$  spread is wall-clock noise, not generation variability. Jobs ran on `ragpu003/ragpu004/ragpu008` (PyTorch 2.5.1+cu124). Llama AWQ-4 and Llama GPTQ-4 have records from more than one hostname.

Output tok/s =  $(\sum \text{output tokens}) / \Delta t_{\text{steady}}$ . GPU-sec/query =  $\Delta t_{\text{steady}} / N$  on one GPU. Peak VRAM is `torch.cuda.max_memory_allocated` after the 0.75 utilization preallocation ( $\sim 60$  GB reserved of 80 GB). Measured

$$C_{\text{pass}}^{\text{meas}} = \frac{(\text{GPU-sec/query}) \times (\$1.50/3600)}{\text{pass@1}}, \quad (1)$$

using the frozen 40-cell MATH-500 means—not accuracy on the 100-item subset. The \$1.50/A100-hour rate is a cloud pricing *scenario*, not a billed cluster price.

### 3.5 Metrics

**Pass@1.** Mean extractive match across seeds. Primary inference is a problem-clustered bootstrap of the paired difference  $\Delta = \overline{\text{acc}}_{\text{quant}} - \overline{\text{acc}}_{\text{BF16}}$  (resample  $n$  items with replacement; retain all seeds per item;  $B = 10,000$ ). We report  $\Delta$  in percentage points, a 95% percentile CI, and a two-sided bootstrap  $p$ -value, with Holm–Bonferroni across the six MATH (resp. GSM8K, GPQA) contrasts. A  $\pm 1$  pp two one-sided tests (TOST) using the 90% CI is reported as an equivalence check, not as a default claim.

**maj@5 McNemar (secondary).** A problem is maj@5-correct if at least 3 of 5 MATH-500 seeds match gold. Exact McNemar tests this consensus, not pass@1.

**Pathology.** A row is a *loop* if the longest consecutive identical-word run is  $\geq 20$  (Unicode word tokens, case-folded). A row is a *cap hit* if re-encoded completion length is  $\geq 32,768$ . Because decode-then-re-tokenize undershoots the cap (observed max 32,737), we also count *near-cap* rows with  $\geq 32,500$  tokens. Saved rows do not include vLLM `finish_reason` or output token IDs; phrase-level cycles are invisible to the identical-word detector.

**Length.** Primary estimator: ratio of means over all seeds, plus the mean paired per-item token delta with a clustered bootstrap CI. We also report the mean of per-item ratios, which is a different estimand and is dominated by short BF16 traces. Length is stratified by the  $2 \times 2$  of BF16/quantized extractive correctness.

**Historical token-implied  $C_{\text{pass}}$ .** Following Erol et al. [3],  $C_{\text{pass}} = (\text{cost per query})/\text{pass@1}$  with

$$\text{Cost per query} = (\bar{T}/\tau) \times (R/3600), \quad (2)$$

$\tau = 65$  tok/s (unmeasured) and  $R = \$1.50$  per A100-hour. Because  $\tau$  and  $R$  are shared, ranking by  $C_{\text{pass}}$  is ranking by  $\bar{T}/\text{pass@1}$ . We retain this proxy only as a sensitivity to the pre-measurement draft; primary systems/cost evidence is the measured GPU-second version in Section 3.4 and Table 8.

**Observable modal agreement.** Compact per-cell JSON stores `extractive_match` but not extracted strings. After recovering the 20,000 MATH-500 answer strings, we form agreement *without* gold. Campaign extraction and clustering used LightEval 0.8.0 with math-verify (frozen in `docs/ANSWER_NORMALIZATION.md`). A later throwaway MacBook install of LightEval 0.8.1 was *not* used for canonical numbers. For item  $i$  and five generated predictions, let  $\sim$  be that campaign mathematical-equivalence relation. Cluster the five predictions into connected components of  $\sim$ . Let  $m_i$  be the size of the unique largest class when one exists; otherwise the item is a non-unique-mode tie. Observable agreement is  $a_i = m_i/5$ . Serve at threshold  $k/5$  only when a unique modal class satisfies  $m_i \geq k$ . The pattern  $(A, A, B, B, C)$  has  $m_i = 2$  and is not served at  $\geq 3/5$ ;  $(A, A, A, B, B)$  has unique mode  $A$  with  $m_i = 3$  and is served at  $\geq 3/5$ . Gold is used only afterward to estimate selective accuracy and risk. All five samples are generated before the decision. The five-sample token-cost proxy  $T_5$  is the sum of their completion tokens, including abstained queries. We report three discrete operating points, not a continuous risk–coverage curve. Gold-hit ECE (confidence =  $c_i/5$ , label =  $\mathbb{I}(c_i \geq 3)$ ) remains a deterministic function of the gold-hit histogram and is omitted.

Table 1: MATH-500 headline grid ( $n = 500$ , seeds 42–46). Clustered 95% CIs resample problems. Loops: identical-word run  $\geq 20$ . Near-cap: completion tokens  $\geq 32,500$ . Exact cap hits after re-encoding: 0 in all cells.

| Cell         | 42   | 43   | 44   | 45   | 46   | Mean $\pm$ Std   | Clust. 95%   | Tok.    | L | NC |
|--------------|------|------|------|------|------|------------------|--------------|---------|---|----|
| Qwen BF16    | 94.4 | 94.0 | 93.8 | 94.6 | 93.2 | $94.00 \pm 0.55$ | [92.2, 95.6] | 4,011.4 | 0 | 14 |
| Qwen FP8     | 94.4 | 95.2 | 94.8 | 92.6 | 95.0 | $94.40 \pm 1.05$ | [92.8, 96.0] | 4,007.7 | 0 | 15 |
| Qwen AWQ-4   | 92.4 | 92.8 | 93.2 | 93.0 | 94.2 | $93.12 \pm 0.67$ | [91.4, 94.8] | 4,265.3 | 1 | 25 |
| Qwen GPTQ-4  | 93.8 | 92.6 | 93.4 | 94.6 | 93.0 | $93.48 \pm 0.77$ | [91.7, 95.2] | 4,287.2 | 0 | 24 |
| Llama BF16   | 89.0 | 88.4 | 90.2 | 89.8 | 88.8 | $89.24 \pm 0.74$ | [87.2, 91.2] | 4,656.6 | 3 | 19 |
| Llama FP8    | 89.0 | 89.6 | 88.6 | 89.2 | 91.2 | $89.52 \pm 1.01$ | [87.4, 91.6] | 4,550.8 | 3 | 10 |
| Llama AWQ-4  | 84.4 | 84.8 | 89.2 | 87.4 | 86.6 | $86.48 \pm 1.96$ | [84.2, 88.7] | 4,736.5 | 3 | 12 |
| Llama GPTQ-4 | 88.0 | 89.6 | 86.8 | 89.4 | 90.8 | $88.92 \pm 1.55$ | [86.9, 90.9] | 4,840.5 | 2 | 21 |

Table 2: Primary: problem-clustered bootstrap of MATH-500 pass@1 versus BF16 ( $B = 10,000$ ). Secondary: exact McNemar on maj@5. Holm at  $\alpha = 0.05$  over six contrasts. TOST  $\pm 1$  pp uses the 90% CI.

| Contrast     | $\Delta$ (pp) | 95% CI         | $p$       | Holm | McNemar $p$ |
|--------------|---------------|----------------|-----------|------|-------------|
| Llama AWQ-4  | -2.76         | [-4.16, -1.44] | $< 0.001$ | sig. | 0.296       |
| Qwen AWQ-4   | -0.88         | [-1.88, +0.08] | 0.069     | n.s. | 1.000       |
| Qwen GPTQ-4  | -0.52         | [-1.40, +0.36] | 0.245     | n.s. | 0.754       |
| Qwen FP8     | +0.40         | [-0.48, +1.28] | 0.383     | n.s. | 0.581       |
| Llama GPTQ-4 | -0.32         | [-1.56, +0.96] | 0.619     | n.s. | 1.000       |
| Llama FP8    | +0.28         | [-0.92, +1.48] | 0.647     | n.s. | 1.000       |

## 4 Results

### 4.1 MATH-500 pass@1

Table 1 and Figure 5 show per-seed pass@1. FP8 point estimates sit 0.28–0.40 pp above BF16. Llama AWQ-4 is the weak MATH cell (−2.76 pp).

Table 2 is the test that matches RQ1. Llama AWQ-4’s clustered interval excludes zero ( $p < 0.001$ ; Holm-significant among the six MATH contrasts). Qwen 4-bit MATH drops are negative but not Holm-significant. No MATH contrast, including FP8, has its 90% CI inside  $[-1, +1]$  pp, so we do *not* claim statistical equivalence at a 1 pp serving margin. Secondary maj@5 McNemar  $p$ -values are all  $> 0.29$ : majority voting absorbs the per-seed flips that pass@1 still sees.

### 4.2 GSM8K and GPQA-Diamond

Table 5: FP8 stays within about one point of BF16 on all three tasks. The largest accuracy effect in the study is Qwen AWQ-4 on GPQA-Diamond (−5.56 pp, 95% CI  $[-9.60, -1.52]$ ,  $p = 0.007$ , Holm-significant among six GPQA contrasts; 3 seeds,  $n = 198$ ). Llama AWQ-4 also drops on GSM8K (−1.57 pp, 95% CI  $[-2.55, -0.58]$ ,  $p = 0.002$ ). Other GSM8K Qwen contrasts pass TOST at  $\pm 1$  pp. GPQA intervals are wide; we treat non-AWQ GPQA contrasts as exploratory.

### 4.3 Loops and near-cap termination

Across 88 cells the validator records **25** repetition-flagged rows (0.044% of 56,408) and **0** exact cap hits. Earlier drafts reported 0/0 because analysis scripts read `truncation_count` / `repetition_flag_count` while the validator writes `token_limit_hits` / `repetition_rows`. Loops occur in BF16 as well as quantized cells (Llama MATH: 3/3/3/2 for BF16/FP8/AWQ/GPTQ).

Table 3: MATH-500 length versus BF16, all five seeds. RoM: ratio of means.  $\Delta$ : mean paired per-item token difference with clustered 95% CI. “Both OK” / “BF16 only”: mean  $\Delta$  on (item, seed) pairs with that correctness pattern.

| Contrast     | RoM   | Mean $\Delta$ [95% CI] | Both OK | BF16 only |
|--------------|-------|------------------------|---------|-----------|
| Qwen FP8     | −0.1% | −4 [−142, +134]        | +51     | +5444     |
| Qwen AWQ-4   | +6.3% | +254 [+116, +398]      | +166    | +6915     |
| Qwen GPTQ-4  | +6.9% | +276 [+121, +431]      | +136    | +6892     |
| Llama FP8    | −2.3% | −106 [−284, +65]       | +38     | +2629     |
| Llama AWQ-4  | +1.7% | +80 [−109, +268]       | +91     | +3090     |
| Llama GPTQ-4 | +3.9% | +184 [+12, +359]       | +66     | +5915     |

Table 4: Historical token-implied Cost-of-Pass on MATH-500 at \$1.50/A100-hour and a shared 65 tok/s. Not measured wall-clock. Ranking equals ranking by mean tokens / pass@1. Primary cost evidence is Table 8.

| Cell         | Mean tok. | Cost / Q  | $C_{\text{pass}}$ |
|--------------|-----------|-----------|-------------------|
| Qwen BF16    | 4,011.4   | \$0.02571 | \$0.02736         |
| Qwen FP8     | 4,007.7   | \$0.02569 | \$0.02721         |
| Qwen AWQ-4   | 4,265.3   | \$0.02734 | \$0.02936         |
| Qwen GPTQ-4  | 4,287.2   | \$0.02748 | \$0.02940         |
| Llama BF16   | 4,656.6   | \$0.02985 | \$0.03345         |
| Llama FP8    | 4,550.8   | \$0.02917 | \$0.03259         |
| Llama AWQ-4  | 4,736.5   | \$0.03036 | \$0.03511         |
| Llama GPTQ-4 | 4,840.5   | \$0.03103 | \$0.03490         |

Three runs exceed 10,000 identical consecutive words. Near-cap rows ( $\geq 32,500$  tokens) total 209; on MATH-500, Qwen AWQ-4/GPTQ-4 have 25 and 24 versus BF16’s 14, and almost all near-cap rows are unboxed. Without `finish_reason` these are inferred length terminations, not proven `length` stops.

#### 4.4 Token length

On the *full* MATH-500 grid and all five seeds (Table 3, Figure 3), Qwen AWQ-4/GPTQ-4 generate +6.33%/+6.88% more tokens than BF16 (mean paired  $\Delta$  +254 and +276 tokens; CIs exclude zero). Llama GPTQ-4 is +3.95% (+184 tokens, CI just excludes zero). Llama AWQ-4 +1.72% and both FP8 cells have CIs that include zero.

A previous 200-item “stratified mixed-correctness” analysis used even indices among the first 400 problems, seed 42 only, and the *mean of per-item ratios*. That estimator is not the full-grid ratio of means: on the same 200 items it reports Qwen FP8 +11.4% while the ratio of means is −3.4%. We discard that subset interpretation.

Figure 4 and Table 3 split paired token deltas by correctness. When both formats are correct, 4-bit adds on the order of +90 to +170 tokens. When BF16 is correct and the quantized sample is wrong, the quantized trace is thousands of tokens longer. Token inflation is therefore largely a *failure-associated* lengthening, which is a more useful reconciliation of Liu et al. [15] (little mean lengthening) and Lian et al. [12] (hidden token cost) than a 200-item even-index slice.

Under the shared- $\tau$  model (Table 4), FP8 has the lowest token-implied  $C_{\text{pass}}$  by \$0.00015 on Qwen. That margin is smaller than seed noise in length. We do not call it Pareto-optimal. Measured batched decode on this stack is an order of magnitude faster than 65 tok/s, so the proxy overstates dollars; relative 4-bit rankings can also flip (Section 4.7).



Table 5: Cross-benchmark pass@1 (mean  $\pm$  seed std). MATH-500: 5 seeds. GSM8K and GPQA-Diamond: 3 seeds. L/NC: loops / near-cap over those seeds.

| Cell         | MATH-500         | GSM8K            | GPQA-D           | L / NC |
|--------------|------------------|------------------|------------------|--------|
| Qwen BF16    | 94.00 $\pm$ 0.55 | 91.26 $\pm$ 0.29 | 50.34 $\pm$ 2.96 | 4 / 28 |
| Qwen FP8     | 94.40 $\pm$ 1.05 | 91.33 $\pm$ 0.16 | 49.49 $\pm$ 1.52 | 0 / 24 |
| Qwen AWQ-4   | 93.12 $\pm$ 0.67 | 91.05 $\pm$ 1.14 | 44.78 $\pm$ 3.04 | 5 / 32 |
| Qwen GPTQ-4  | 93.48 $\pm$ 0.77 | 91.13 $\pm$ 0.27 | 47.98 $\pm$ 1.75 | 3 / 40 |
| Llama BF16   | 89.24 $\pm$ 0.74 | 88.68 $\pm$ 0.46 | 46.13 $\pm$ 1.91 | 3 / 21 |
| Llama FP8    | 89.52 $\pm$ 1.01 | 88.80 $\pm$ 0.62 | 47.81 $\pm$ 0.29 | 3 / 12 |
| Llama AWQ-4  | 86.48 $\pm$ 1.96 | 87.11 $\pm$ 0.23 | 46.97 $\pm$ 2.02 | 4 / 21 |
| Llama GPTQ-4 | 88.92 $\pm$ 1.55 | 88.96 $\pm$ 0.70 | 44.95 $\pm$ 4.32 | 3 / 31 |

Table 6: MATH-500 difficulty labels (5-seed pass@1 mean  $\pm$  std). Labels are not monotone: Level 2 is harder than Level 5 for both BF16 backbones. Level-1  $n = 43$ ; treat cell-wise gaps as descriptive.

| Level           | Qwen BF16        | Qwen FP8         | Qwen AWQ-4       | Qwen GPTQ-4      |
|-----------------|------------------|------------------|------------------|------------------|
| 1 ( $n = 43$ )  | 98.60 $\pm$ 1.27 | 98.60 $\pm$ 2.08 | 98.14 $\pm$ 1.95 | 96.28 $\pm$ 2.08 |
| 2 ( $n = 90$ )  | 89.11 $\pm$ 0.93 | 90.44 $\pm$ 4.20 | 90.67 $\pm$ 1.69 | 88.67 $\pm$ 2.14 |
| 3 ( $n = 105$ ) | 95.81 $\pm$ 1.09 | 95.24 $\pm$ 1.51 | 94.10 $\pm$ 0.80 | 94.29 $\pm$ 1.90 |
| 4 ( $n = 128$ ) | 95.78 $\pm$ 0.89 | 96.41 $\pm$ 1.42 | 94.53 $\pm$ 1.24 | 96.56 $\pm$ 1.18 |
| 5 ( $n = 134$ ) | 92.69 $\pm$ 1.11 | 93.13 $\pm$ 1.23 | 91.04 $\pm$ 1.67 | 92.24 $\pm$ 1.95 |
| Level           | Llama BF16       | Llama FP8        | Llama AWQ-4      | Llama GPTQ-4     |
| 1 ( $n = 43$ )  | 93.95 $\pm$ 3.12 | 94.88 $\pm$ 3.03 | 95.35 $\pm$ 1.64 | 96.28 $\pm$ 2.65 |
| 2 ( $n = 90$ )  | 84.67 $\pm$ 2.14 | 85.11 $\pm$ 2.02 | 81.11 $\pm$ 3.04 | 83.33 $\pm$ 3.42 |
| 3 ( $n = 105$ ) | 89.90 $\pm$ 2.48 | 90.10 $\pm$ 1.73 | 85.71 $\pm$ 1.17 | 89.71 $\pm$ 1.83 |
| 4 ( $n = 128$ ) | 91.56 $\pm$ 1.69 | 91.09 $\pm$ 0.43 | 87.03 $\pm$ 2.25 | 89.84 $\pm$ 1.75 |
| 5 ( $n = 134$ ) | 88.06 $\pm$ 1.75 | 88.81 $\pm$ 2.84 | 87.31 $\pm$ 3.46 | 88.81 $\pm$ 1.18 |

#### 4.5 Difficulty labels

Table 6: Level 1 is easy for every format in this sample. Llama AWQ-4 is weakest on Levels 2–4. MATH level labels are not a monotone difficulty axis here (Level 2 Qwen BF16 89.11% vs Level 5 92.69%).

#### 4.6 Observable modal agreement on MATH-500

Recovered answer strings make RQ3 measurable as observable multi-sample agreement, selective prediction, and abstention—not as model safety or softmax calibration. Table 7 reports coverage and selective risk at three discrete thresholds. On MATH-500, higher observable answer agreement is associated with lower selective error. At 5/5 consensus, selective error is at most 0.27% across the evaluated configurations, but coverage ranges from 70.2% (Llama AWQ-4) to 88.8% (Qwen FP8). Point estimates of 100% selective accuracy among served items are finite-sample statements and do not imply zero true error or guaranteed correctness.

AWQ-4 reduces strict-consensus coverage in some configurations, most clearly for Llama-8B: 70.2% versus 76.2% for BF16, a paired difference of  $-6.0$  pp (95% CI  $[-9.4, -2.6]$ ). Llama AWQ-4  $\geq 4/5$  coverage falls 3.2 pp ( $[-6.0, -0.6]$ ). Qwen AWQ-4  $\geq 4/5$  coverage falls 2.2 pp ( $[-4.0, -0.6]$ ). The  $\geq 3/5$  AWQ contrasts, and all GPTQ-4 coverage contrasts, have paired intervals that include zero. No clear FP8–BF16 difference was detected in modal-agreement coverage or selective risk under these configurations. Appendix Table 9 lists the full threshold pattern.

Table 7: Observable modal-answer agreement on MATH-500 ( $n = 500$ , seeds 42–46). Coverage is the fraction of items with a unique modal class of size  $\geq k$ . Risk is 1–selective accuracy among served items. Mean  $T_5$  is the five-sample token-cost proxy over all 500 items (abstained queries still pay). Gold is used only to score served predictions.

| Model    | Fmt    | $\geq 3/5$ cov. | $\geq 3/5$ risk | $\geq 4/5$ cov. | $\geq 4/5$ risk | $5/5$ cov. | $5/5$ risk | Mean $T_5$ |
|----------|--------|-----------------|-----------------|-----------------|-----------------|------------|------------|------------|
| Qwen-7B  | BF16   | 96.0            | 1.67            | 93.4            | 0.43            | 88.4       | 0.23       | 20,057     |
| Qwen-7B  | FP8    | 96.6            | 1.66            | 92.6            | 0.00            | 88.8       | 0.00       | 20,038     |
| Qwen-7B  | AWQ-4  | 95.8            | 1.46            | 91.2            | 0.44            | 86.4       | 0.23       | 21,327     |
| Qwen-7B  | GPTQ-4 | 95.4            | 1.47            | 92.8            | 0.43            | 86.8       | 0.00       | 21,436     |
| Llama-8B | BF16   | 94.0            | 2.98            | 87.8            | 0.68            | 76.2       | 0.26       | 23,283     |
| Llama-8B | FP8    | 94.0            | 3.19            | 88.6            | 0.90            | 78.2       | 0.00       | 22,754     |
| Llama-8B | AWQ-4  | 93.2            | 3.65            | 84.6            | 1.89            | 70.2       | 0.00       | 23,682     |
| Llama-8B | GPTQ-4 | 93.8            | 2.77            | 86.4            | 0.46            | 75.4       | 0.27       | 24,203     |

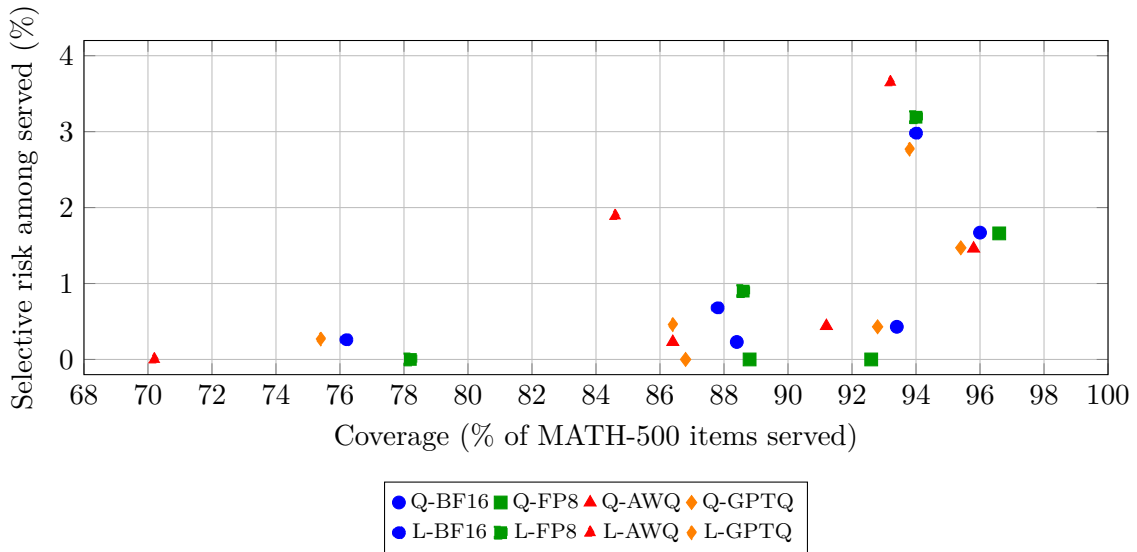


Figure 1: Three discrete MATH-500 operating points ( $\geq 3/5$ ,  $\geq 4/5$ ,  $5/5$ ), not a continuous risk–coverage curve. Solid markers: Qwen-7B. Dashed markers: Llama-8B. Lower-right is high coverage and higher residual risk; lower-left is strict consensus.

The five-sample token-cost proxy is paid before abstention. Mean  $T_5$  is 20,057 / 20,038 / 21,327 / 21,436 tokens for Qwen BF16/FP8/AWQ-4/GPTQ-4 and 23,283 / 22,754 / 23,682 / 24,203 for Llama. Qwen 4-bit retains its approximately +6–7% output-token penalty in this accounting because  $T_5$  is the sum of the same five traces. Tokens per served query divide that total by the number served, so abstained items are not free. This is not a measured monetary cost. Figure 1 shows the three operating points.

#### 4.7 Measured serving throughput and Cost-of-Pass

Table 8 reports the batched condition (100 prompts, continuous batching). Qwen-7B FP8 is +18.7% output tok/s versus BF16 ( $824.27 \pm 4.00$  vs.  $694.27 \pm 0.65$ ) and  $-19.8\%$  measured  $C_{\text{pass}}$  (\$0.0020 vs. \$0.0025) at \$1.50/A100-hour. Llama-8B FP8 is +16.9% tok/s and  $-7.4\%$   $C_{\text{pass}}$ . These FP8 numbers are A100 Marlin W8A16, not Hopper W8A8.

Four-bit speed is not one number. Batched Llama GPTQ-4 reaches  $785.55 \pm 12.33$  tok/s (+20.9% vs. Llama BF16). Qwen GPTQ-4 is slower than Qwen BF16 ( $-13.2\%$ ). Llama AWQ-4 batched mean  $545.68 \pm 125.62$  tok/s includes a same-token, slower third repeat on a different



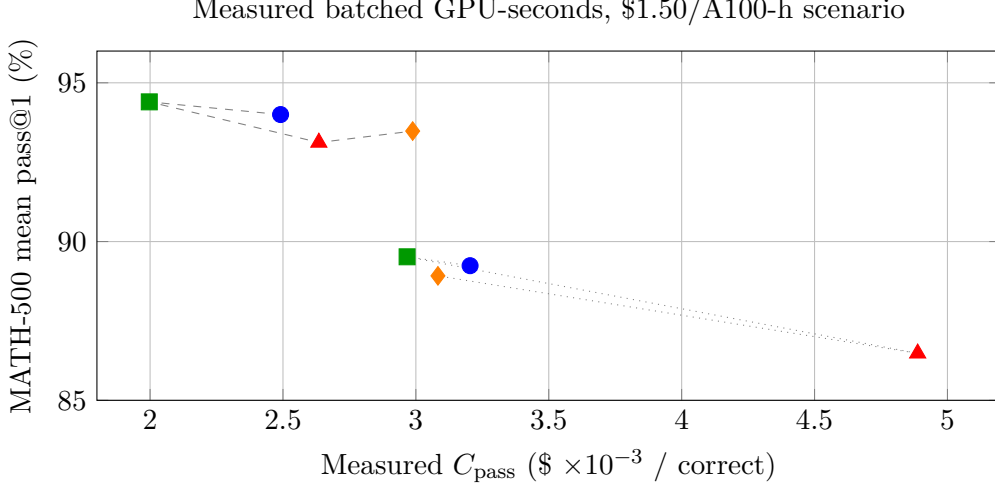


Figure 2: Measured batched  $C_{\text{pass}}$  versus campaign MATH-500 pass@1 ( $\$ \times 10^{-3}$ ). Solid/dashed lines are Qwen/Llama visual guides, not a fitted frontier. Nondominated on (pass@1,  $C_{\text{pass}}$ ): Qwen-7B FP8 (pooled); Llama-8B FP8 (within Llama). Not a unique “true Pareto optimum.”

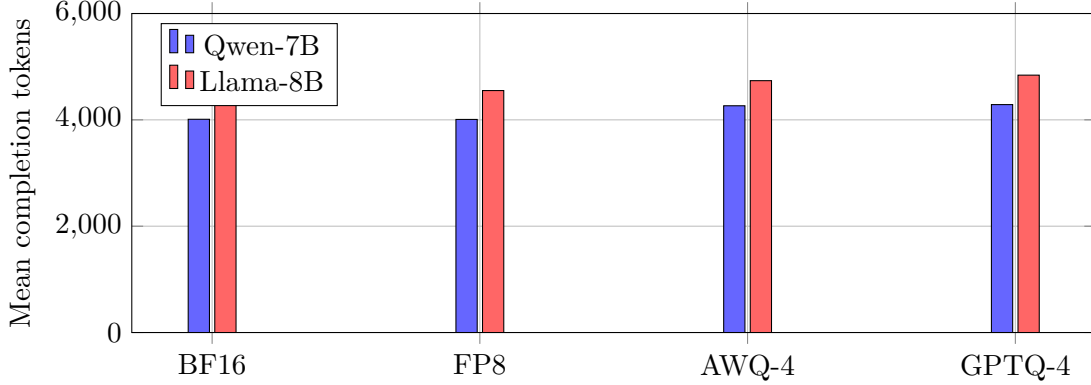


Figure 3: Full-grid mean MATH-500 completion tokens (ratio-of-means estimand).

node (368 tok/s vs.  $\sim 634$  tok/s on the other two); we report the protocol  $R = 3$  mean and do not drop the outlier. Single-stream 4-bit is uniformly slower than matched BF16 (Appendix Table 10): Qwen AWQ-4  $-8.9\%$ , Qwen GPTQ-4  $-14.5\%$ , Llama AWQ-4  $-5.4\%$ , Llama GPTQ-4  $-2.5\%$  tok/s. That is Marlin dequant overhead at concurrency 1, not a memory-capacity result.

Token inflation does not automatically cancel a kernel speedup. On the batched subset, Qwen AWQ-4 is  $+3.8\%$  mean output tokens vs. BF16 and  $-1.0\%$  tok/s, so GPU-sec/query still rises  $+4.8\%$ . Qwen GPTQ-4 is  $+3.6\%$  tokens and  $-13.2\%$  tok/s. Llama GPTQ-4 is  $+15.9\%$  tokens but  $+20.9\%$  tok/s (time ratio 0.96). Cost is tokens  $\times$  cost-per-token; neither factor alone is the ranking.

Peak allocated VRAM is 55.2–56.7 GB for every batched cell. That is the 0.75 utilization pool, not a measurement of 4-bit weight-footprint savings. Memory and latency are separate.

On (pass@1, measured  $C_{\text{pass}}$ ) over the eight batched points, the nondominated set is {Qwen-7B FP8}. Within Llama the nondominated point is Llama FP8. Qwen GPTQ-4 is dominated. We do not call any cell a “true Pareto optimum”: the frontier is two-objective, scenario-priced, and stack-specific (Figure 2). The old 65 tok/s proxy (Table 4) remains a sensitivity, not the primary cost evidence.

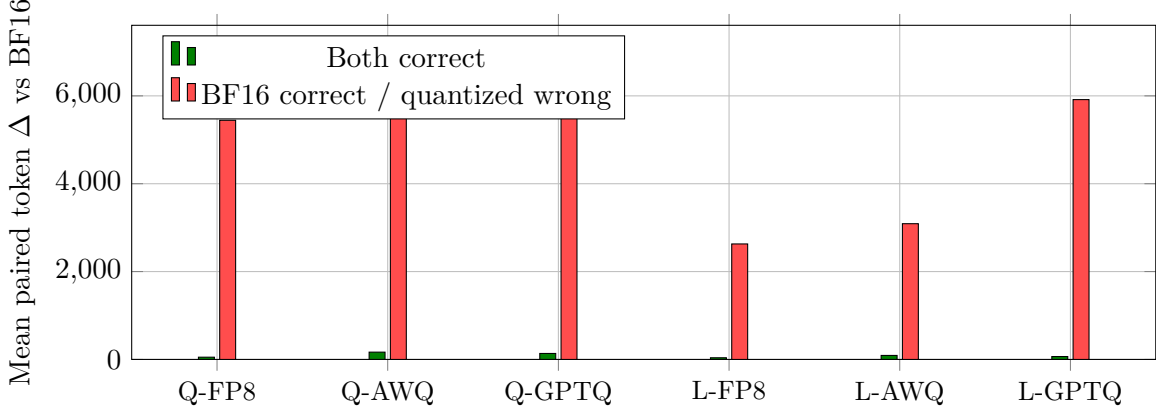


Figure 4: Paired token deltas by correctness. Extra length is concentrated on format-induced failures, not on uniformly longer correct traces.

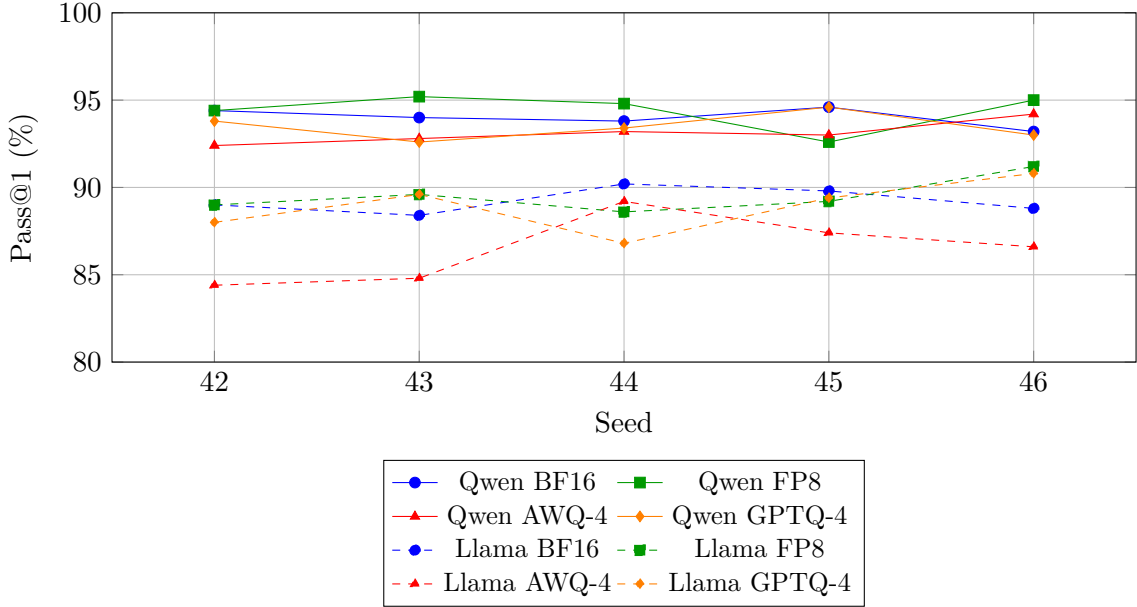


Figure 5: MATH-500 pass@1 by seed. Solid: Qwen-7B. Dashed: Llama-8B. Llama AWQ-4 is the unstable 4-bit cell.

Table 8: Measured batched serving (Condition B: 100-prompt `llm.generate`,  $R = 3$  wall-clock repeats). Output tok/s is mean  $\pm$  population std. GPU-sec/query =  $\Delta t / N$  on one A100.  $C_{\text{pass}}$  uses campaign MATH-500 pass@1 and \$1.50/A100-hour. Peak VRAM is allocated bytes after `gpu_memory_utilization=0.75`. Llama AWQ-4 std is inflated by a mixed-node repeat (see text).

| Model    | Fmt    | tok/s               | req/s | GPU-s/q | VRAM (GB) | $C_{\text{pass}}$ |
|----------|--------|---------------------|-------|---------|-----------|-------------------|
| Qwen-7B  | BF16   | 694.27 $\pm$ 0.65   | 0.178 | 5.62    | 56.00     | \$0.0025          |
| Qwen-7B  | FP8    | 824.27 $\pm$ 4.00   | 0.221 | 4.52    | 56.02     | \$0.0020          |
| Qwen-7B  | AWQ-4  | 687.51 $\pm$ 0.59   | 0.170 | 5.89    | 56.00     | \$0.0026          |
| Qwen-7B  | GPTQ-4 | 602.65 $\pm$ 1.40   | 0.149 | 6.70    | 55.17     | \$0.0030          |
| Llama-8B | BF16   | 649.72 $\pm$ 0.51   | 0.146 | 6.86    | 56.71     | \$0.0032          |
| Llama-8B | FP8    | 759.48 $\pm$ 6.96   | 0.157 | 6.37    | 56.72     | \$0.0030          |
| Llama-8B | AWQ-4  | 545.68 $\pm$ 125.62 | 0.105 | 10.15   | 56.72     | \$0.0049          |
| Llama-8B | GPTQ-4 | 785.55 $\pm$ 12.33  | 0.152 | 6.58    | 56.74     | \$0.0031          |

## 5 Discussion

Under this pinned A100 W8A16 stack, FP8-checkpoint serving tracks BF16 pass@1 closely on the evaluated distillations, but we do not establish  $\pm 1$  pp equivalence on MATH-500. Four-bit behavior is not a single story: Qwen keeps MATH-500 and GSM8K near BF16 while losing 5.6 pp on GPQA-Diamond under AWQ-4; Llama AWQ-4 is the MATH/GSM8K weak cell. Mean 4-bit lengthening on MATH-500 is modest and, when present, lives mostly on traces that also lose the answer. That is a sharper claim than “4-bit inflates reasoning” and a more honest one than “nothing happens.”

Pass@1 and stochastic output stability are related but distinct. Llama AWQ-4’s degradation appears in both mean accuracy and 5/5 consensus coverage. Qwen 4-bit formats retain relatively strong MATH-500 pass@1 while spending more tokens. No clear FP8–BF16 difference was detected in modal-agreement coverage or selective risk under these configurations; paired intervals that include zero do not prove equivalence. Repeated sampling sharply reduces selective risk on MATH-500 but costs about five generations per attempted item. The construction is not a general uncertainty estimator and is not a safety property; it is an abstention rule on MATH-500 that requires five generations before the decision.

We previously wrote that pinning vLLM 0.7.0 “removed” truncations and loops seen on vLLM 0.8.5. Those earlier runs also differed in harness, decoding flags (including a bug that dropped `repetition_penalty`), and prompt profile. The final grid still contains 25 loops. We therefore do not attribute pathology rates causally to the engine version. A clean stack  $\times$  format factorial remains future work, not a result in this paper.

FP8 is a reasonable *accuracy* default on this A100 W8A16 path for these two distillations. On the measured batched subset it is also the lowest  $C_{\text{pass}}$  point in the eight-cell (pass@1, cost) set, at +18.7% Qwen tok/s versus BF16. It is still not Hopper native W8A8, and single-stream FP8 does not show the same cost cut (Qwen FP8  $C_{\text{pass}}$  +1.0% vs. BF16 at  $C = 1$ ).

Cost is tokens and cost-per-token. Qwen 4-bit lengthening on the 56k MATH-500 grid is real; on the serving subset it does not, by itself, decide dollars. Batched 4-bit kernels can raise tok/s (Llama GPTQ-4 +20.9%) or not (Qwen GPTQ-4 −13.2%). Single-stream 4-bit is slower than BF16 on every cell we timed: dequant overhead dominates when the GPU is not batch-saturated. Peak allocated VRAM does not separate formats under a 0.75 utilization cap; do not read Table 8 as a memory-compression result.

## 6 Limitations

1. **A100 W8A16 fallback.** FP8 numbers do not transfer to Hopper native W8A8.
2. **Measured serving scope.** Throughput, latency, GPU-sec/query, and scenario  $C_{\text{pass}}$  are measured on a 100-problem MATH-500 subset (20 sequential prompts for Condition A), not the full 56k campaign. Pass@1 in  $C_{\text{pass}}$  is the 40-cell campaign mean. The \$1.50/A100-hour rate is a pricing scenario. Condition B is a 100-prompt `llm.generate` (continuous batching; `max_num_seqs` not pinned to 8). Condition A first-20 level counts are 5/7/3/3/2. Repetitions share sampling seed 20260816. Llama AWQ-4/GPTQ-4 records mix host-names. Peak allocated VRAM is the 0.75 engine pool, not isolated weight footprint. Energy is not measured.
3. **Pathology metrology.** The loop detector is identical-word runs  $\geq 20$ . Phrase-level cycles and `finish_reason` were not saved in the original campaign. Near-cap is a proxy.
4. **Modal agreement.** Five generations are required per attempted MATH problem. Only MATH-500 has five seeds in this campaign. Mathematical answer equivalence is evaluator- and task-specific (campaign LightEval 0.8.0 / math-verify). High agreement can still be

wrong. The signal is not general uncertainty calibration, not softmax calibration, and not a safety property.

5. **Checkpoint provenance.** AWQ-4 is a community quantization of unknown calibration set and quantizer revision. GPTQ-4/FP8 are RedHatAI artifacts (appendix).
6. **Scope.** Two 7–8B DeepSeek-R1 distillations, English math/science, one pinned serving stack, one prompt. GSM8K/GPQA use three seeds.
7. **Source configs vs. launcher.** Files in `configs/models/` are not the `QRM inference.py` campaign. Effective KV-cache dtype is not stored per cell. Launcher flags such as `max_model_len=32768` are code/launcher-confirmed, not per-job runtime telemetry.

## 7 Conclusion

On a pinned vLLM 0.7.0 eager stack, FP8-checkpoint (W8A16) pass@1 stays close to BF16 for two R1 distillations, while 4-bit effects are model- and task-dependent and are visible to clustered pass@1 tests that maj@5 McNemar misses. Loops are rare but nonzero; near-cap terminations are more common for Qwen 4-bit on MATH-500. Mean token inflation, where it exists, is concentrated on failures. Observable 5-sample agreement on MATH-500 yields low selective error at strict consensus, at the cost of five generations, and Llama AWQ-4 reduces 5/5 coverage relative to BF16. Measured batched serving makes Qwen FP8 both faster (+18.7% tok/s) and cheaper (−19.8%  $C_{\text{pass}}$ ) than Qwen BF16 under a \$1.50/A100-hour scenario; 4-bit speed depends on concurrency and kernel, and allocated VRAM does not separate formats. The contribution is a stack-pinned measurement of those effects, not a new compressor and not a serving-stack shift.

## Artifacts

Code, patches, validation JSON, and analysis reports: [Manish06N/reasoning-compression-lab](https://github.com/Manish06N/reasoning-compression-lab). Per-cell records: `results/math500/`, `results/gsm8k/`, `results/gpqa/`. Canonical pass@1 tables: `revision_reanalysis_report.json`. Modal agreement: `modal_agreement_report.json`. JSON rows contain `extractive_match`, `completion_tokens`, `repetition_flag`; they do *not* contain traces, token IDs, or `finish_reason`. See `paper/ARTIFACT.md`. GPQA is a gated benchmark; public records must not republish item text.

## Acknowledgment

PARAM Rudra HPC at IIT Patna (National Supercomputing Mission, C-DAC) provided compute.

## A Prompt

MATH-500 and the GSM8K/GPQA QRM templates use:

```
{question}
```

```
Please reason step by step, and put your final  
answer within \boxed{}
```

with a `<think>` prefix on the official QRM path.

## B Checkpoints

| Cell         | Hugging Face ID / revision                                                                        |
|--------------|---------------------------------------------------------------------------------------------------|
| Qwen BF16    | deepseek-ai/DeepSeek-R1-Distill-Qwen-7B<br>916b56a44061fd5cd7d6a8fb632557ed4f724f60               |
| Qwen FP8     | RedHatAI/DeepSeek-R1-Distill-Qwen-7B-FP8-dynamic<br>ceb2fcd178d477cd21b92ffb43164c74165a212c      |
| Qwen AWQ-4   | jakiAJK/DeepSeek-R1-Distill-Qwen-7B_AWQ<br>1d082b3ded4e7cefb6b32cd5e07c7d409832816                |
| Qwen GPTQ-4  | RedHatAI/DeepSeek-R1-Distill-Qwen-7B-quantized.w4a16<br>9559a91ed463eecd64b93a532fe5aa8cc25f2eb   |
| Llama BF16   | deepseek-ai/DeepSeek-R1-Distill-Llama-8B<br>6a6f4aa4197940add57724a7707d069478df56b1              |
| Llama FP8    | RedHatAI/DeepSeek-R1-Distill-Llama-8B-FP8-dynamic<br>5d548d9169c53b7bb7ef0b3bc261509d5da6e3dd     |
| Llama AWQ-4  | jakiAJK/DeepSeek-R1-Distill-Llama-8B_AWQ<br>57360e84a0e7278c6b3abb453f2dc812b213badb              |
| Llama GPTQ-4 | RedHatAI/DeepSeek-R1-Distill-Llama-8B-quantized.w4a16<br>faa06ed0a4a5bdab76c4df12376e0c5228c0bdfe |

AWQ-4 checkpoints are third-party community uploads; calibration corpus and quantization code revision are unknown. GPTQ-4 and FP8 are RedHatAI compressed-tensors artifacts. Dataset revision SHAs for MATH-500 / GSM8K / GPQA-Diamond were not pinned in the campaign launcher beyond the QRM task names.

Serving protocol and raw JSON live under `docs/MEASURED_SERVING_PROTOCOL.md` and `results/measured_serving/raw/`. Aggregates: `measured_serving_report.json`.

## C Reanalysis

Pathology, clustered bootstrap, and token strata are produced by the canonical reanalysis script from the released per-cell JSON. Loop threshold = 20. Near-cap threshold = 32,500. Bootstrap  $B = 10,000$ , seed 0.

## D Modal-agreement details

The compact recovered artifact has 20,000 rows (SHA256 prefix 23e9ead021111959; full digest is in the JSON sidecar). It stores extracted prediction representations, campaign `extractive_match`, and completion tokens. It does not store chain-of-thought or problem text. Campaign extraction used LightEval 0.8.0; a later MacBook LightEval 0.8.1 install was not used for these numbers. Campaign extractive match was reproduced on 20,000/20,000 rows before clustering. Equivalence clustering reported 0 symmetry violations, 0 transitivity violations, and 116 non-unique-mode ties among 4,000 groups. Bootstrap uses  $B = 10,000$  problem resamples with seed 20260816. Secondary ECE/Brier scores (confidence =  $m_i/5$ ) are in the JSON report and are not treated as a primary calibration result.

$T_5$  is the five-sample token-cost proxy: the sum of completion tokens over seeds 42–46. Mean / median / P90  $T_5$  are computed over all 500 items. Tokens per served query and tokens per correct served result use that same 500-item  $T_5$  total in the numerator, so abstained queries are not free.

## E Measured serving: single-stream and microbenchmark

Condition A (Table 10) is sequential decode of the first 20 prompts of the frozen 100-item list. Median latency is the mean of the three per-run medians. Fixed-token microbenchmark

Table 9: Paired coverage  $\Delta$  versus matched BF16 (percentage points). Intervals that exclude zero are marked \*. Selective-accuracy  $\Delta$  intervals are in the machine-readable modal report.

| Contrast     | $\geq 3/5$ cov. $\Delta$ [95% CI] | $\geq 4/5$ cov. $\Delta$ [95% CI] | $5/5$ cov. $\Delta$ [95% CI] |
|--------------|-----------------------------------|-----------------------------------|------------------------------|
| Qwen FP8     | +0.6 [−0.8, +2.0]                 | −0.8 [−2.2, +0.6]                 | +0.4 [−1.6, +2.4]            |
| Qwen AWQ-4   | −0.2 [−1.8, +1.2]                 | −2.2 [−4.0, −0.6]*                | −2.0 [−4.4, +0.2]            |
| Qwen GPTQ-4  | −0.6 [−2.0, +0.8]                 | −0.6 [−2.2, +1.0]                 | −1.6 [−4.2, +0.8]            |
| Llama FP8    | −0.0 [−1.8, +1.8]                 | +0.8 [−1.6, +3.2]                 | +2.0 [−1.2, +5.2]            |
| Llama AWQ-4  | −0.8 [−2.8, +1.2]                 | −3.2 [−6.0, −0.6]*                | −6.0 [−9.4, −2.6]*           |
| Llama GPTQ-4 | −0.2 [−2.2, +1.8]                 | −1.4 [−3.8, +1.0]                 | −0.8 [−4.2, +2.6]            |

Table 10: Measured single-stream serving (Condition A: 20 sequential prompts,  $R = 3$ ). Latency columns are means of per-run median/P90.  $C_{\text{pass}}$  uses campaign MATH-500 pass@1 and \$1.50/A100-hour.

| Model    | Fmt    | tok/s            | Med. lat (s) | P90 lat (s) | GPU-s/q | $C_{\text{pass}}$ |
|----------|--------|------------------|--------------|-------------|---------|-------------------|
| Qwen-7B  | BF16   | 79.91 $\pm$ 0.07 | 32.97        | 77.76       | 45.44   | \$0.0201          |
| Qwen-7B  | FP8    | 85.89 $\pm$ 0.79 | 41.87        | 69.14       | 46.07   | \$0.0203          |
| Qwen-7B  | AWQ-4  | 72.76 $\pm$ 0.16 | 51.47        | 174.46      | 73.96   | \$0.0331          |
| Qwen-7B  | GPTQ-4 | 68.34 $\pm$ 0.66 | 46.20        | 81.85       | 48.38   | \$0.0216          |
| Llama-8B | BF16   | 73.41 $\pm$ 0.70 | 47.66        | 192.52      | 74.79   | \$0.0349          |
| Llama-8B | FP8    | 80.44 $\pm$ 1.22 | 40.23        | 113.75      | 60.87   | \$0.0283          |
| Llama-8B | AWQ-4  | 69.43 $\pm$ 2.09 | 49.00        | 212.65      | 79.97   | \$0.0385          |
| Llama-8B | GPTQ-4 | 71.60 $\pm$ 0.96 | 59.53        | 227.90      | 99.00   | \$0.0464          |

(Table 11): 10 prompts  $\times$  512 tokens,  $T = 0.0$ , `ignore_eos`. Llama AWQ-4 microbenchmark (438 tok/s) ran on `ragpu003`, the same slower node as the batched outlier.



Table 11: Fixed-token decode microbenchmark (10 prompts  $\times$  512 tokens). Isolates kernel decode speed from trace-length variation.

| Model    | Fmt    | Raw decode tok/s |
|----------|--------|------------------|
| Qwen-7B  | BF16   | 724.67           |
| Qwen-7B  | FP8    | 787.55           |
| Qwen-7B  | AWQ-4  | 678.55           |
| Qwen-7B  | GPTQ-4 | 608.84           |
| Llama-8B | BF16   | 709.73           |
| Llama-8B | FP8    | 761.49           |
| Llama-8B | AWQ-4  | 437.58           |
| Llama-8B | GPTQ-4 | 743.30           |

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